

Unsupervised KMeans Clustering of Weather Data for Storm Identification

Final project report

Introduction to Machine Learning for Physical Sciences

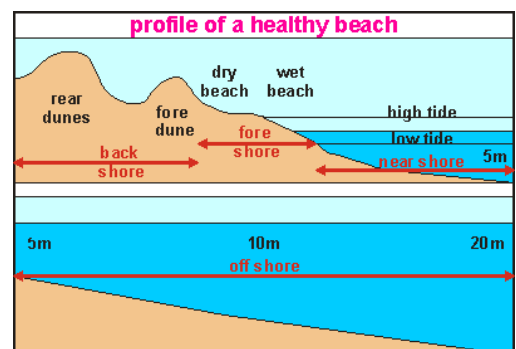
By Max Callahan

Introduction

For my master research project in the Geography department, I am exploring a method to derive the width of beaches from high resolution satellite images. As part of my research, we have been conducting quarterly surveys of twelve beaches in California, gathering vegetation transect data, precision RTK surveys of beach height profiles, and flying drone missions to gather high resolution imagery of each beach as well as digital elevation maps of the whole beach site corrected with our ground based RTK measurements. A central mission of my personal research is to increase the spatial scale and replicability of these labor intensive field surveys with satellite based methods to study sandy beach and dune dynamics. This would allow us to study much larger swathes of the California coast and study beach morphology throughout time across many sites. An accomplishable metric I have landed on to study this morphology is the width of the Dry Upper Beach, the section of dry sand between the toe of the dunes and the high tide line. This beach feature is a great indicator of the total sand budget of the dune system, the health of the beach, and the seaward erosion or accretion of dunes. I have created a time series for the width of Samoa beach in Humboldt County, California, using a semi automated edge detection method by delineating the edge of the dunes and the high tide extent in 3 meter planetscope imagery. Taking this dataset, I want to further understand the climatic drivers of beach erosion on Samoa beach, and build a framework for further analysis as I gather more beach widths across the state. For this project, I have developed a methodology for detecting storm-driven erosion events using historical weather data from a variety of sources close to my site using KMeans clustering. This has improved the replicability of my storm detection from raw weather data, and is a robust methodology I will continue to apply in my research.

Background

The Dry Upper Beach is a uniquely important and ecologically distinct environment in the local beach spatial scale. Beaches represent the edges of two distinct worlds, the land and the ocean, and represent the stage on which vital cross-ecosystem interactions move nutrients from the ocean onto land, and vice versa. The dry upper beach is a central space in these interactions, hosting a unique variety of shorebird habitat, invertebrates, and plant species. Kelp wrack, broken free from rocky coastal reefs, is deposited on the beach yearly. The majority of the fraction of this land-deposited kelp ends up on our sandy beaches, where invertebrates and insects decompose the kelp tangles, nourishing the sand with nitrogen for native dune species, as well as feeding this important basis of the beach food web. Shorebirds, foxes, and other small animals feed on these



invertebrates, moving this energy source inland to many ecosystems across California.¹

But this delicate ecosystem is under the most pressing threat from Global Sea Level Rise (GSLR) and increased storm surge in the face of a changing ocean climate. In the following images we can see the



results of the 2024-2025 winter storm season on Samoa beach, calculated from our ground based RTK measurements, the dune toe eroded nearly 20 meters, while the elevation of the beach fell nearly 3 meters. As can be seen from the images, the high tide lines between the two images show the dramatic reduction in the availability of dry sand, squeezing this vital habitat to essentially nothing within a single winter. Note the absence of kelp wrack, driftwood, and other debris is the eroded beach state. This is due to the water constantly inundating this site with wave runup, eliminating the dry upper beach habitat. This makes the large-scale study of California's beach widths a vital strategy in the management and conservation of these delicate ecosystems and the vital interactions they support.

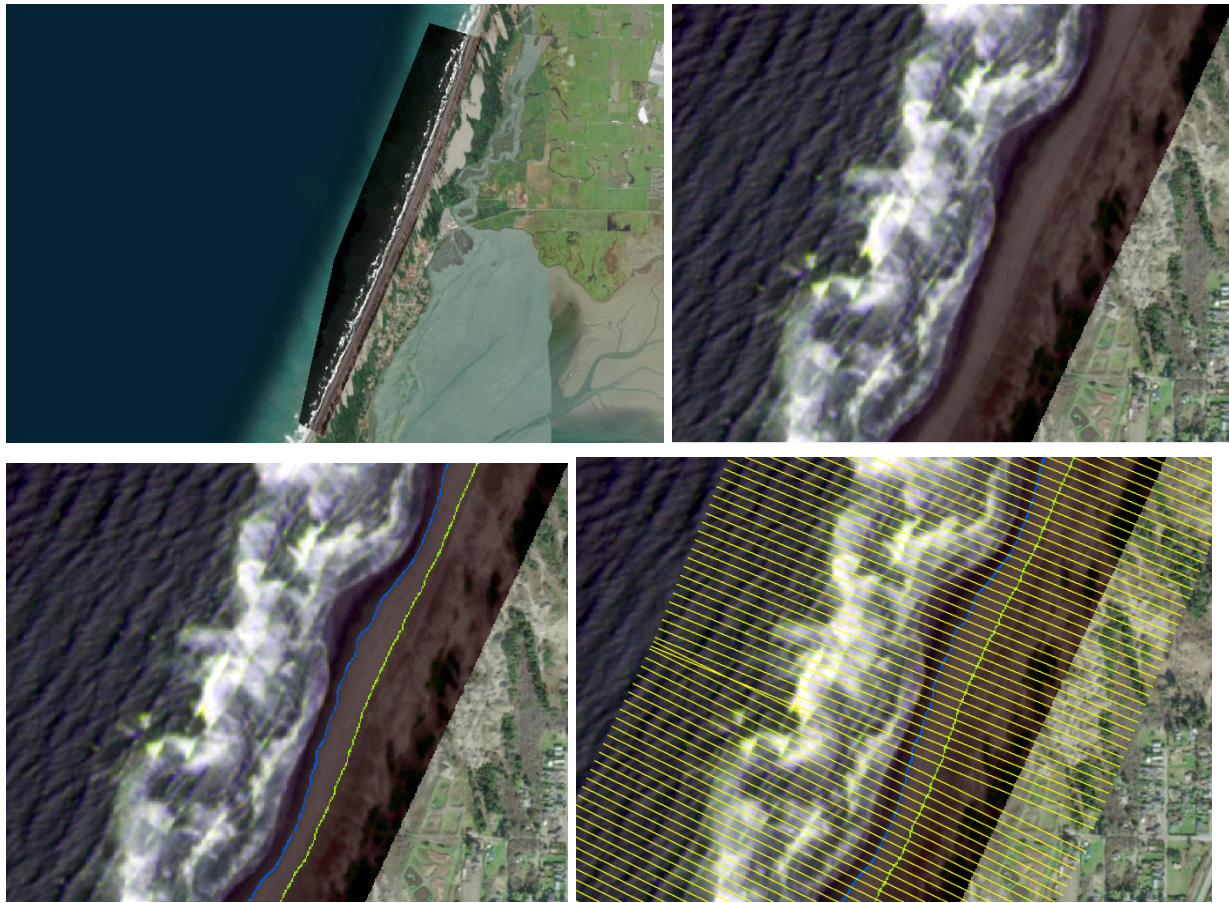
Methods

Downloading satellite imagery and edge detection

Satellite imagery for the site is first downloaded from Planet Labs using their API which I customized to remove the tidal signal. I did this by downloading the tide information from the site, and then filtering only for dates where the median tide height fell within the satellite pass over the region. Planet Lab's constellation of satellites delivers near daily images at a similar time each day (+/- 1 hr) making this an easier task. I then downloaded only images of this mid-tide for my time series analysis, filtering for clouds and bad imagery. This resulted in around 2 days per month of imagery for my time series (2017 - 2024). The delineation of the beach width is first derived using Canny edge detection in Matlab, creating a binary raster which identifies the strong edges in the image. This method works well to delineate the wet-dry line in the sand, as well as the seaward edge of dune vegetation, two essential features in studying the width of the dry upper beach. I then filter first by length (keeping only the most prominent and longest line sections) and then finally manually edit out the erroneously classified lines in the image, to leave only the classified beach features, tide-line and dune-toe. This left me with a time series of these features from

¹ Emery, K.A., Dugan, J.E., Hubbard, D.M. *et al.* Food web structure and ecosystem multifunctionality in a subsidized coastal ecosystem. *Sci Rep* 15, 39086 (2025). <https://doi.org/10.1038/s41598-025-25395-5>

2017 to 2024 with 70 distinct dates (some months only had one suitable satellite image, and a few did not have any).



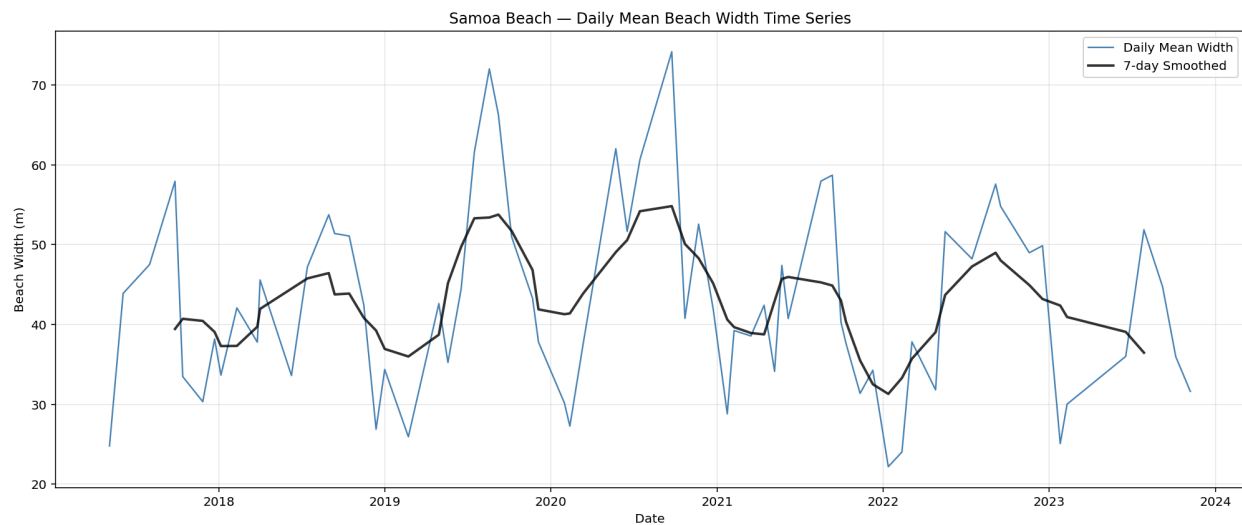
Top left: raw satellite image, top right: zoomed raw image to show tide line and dune field, bottom left: postprocessed Canny detected edges, bottom right: generated transects

Making transects and pulling widths

Once the tide and dune toe lines are created for every date, I created a script which creates evenly spaced transects along a reference line. This is done by inputting parameters such as the spacing between the transects (I made transects every 20 meters), and the length of each transect segment (100 on each side of the line, so 200 total meters). The script then iterates over the reference line, and every 20 meters finds the tangent to the reference and creates a perpendicular line, thereby able to account for the curvature of the beach through the curvature of the reference line. This resulted in the creation of 398 transects along the study site. Next I took the point to point euclidean distance in the UTM zone of the site (UTM zone 10 for Samoa beach) to get an output line distance between the tide line feature and the dune toe created before. The resulting outputs of beach width were saved in a csv containing: transect ID (1-398), date, and width in meters.

Plotting the time series

I then plotted the average width between the transects for each date across the whole time series to observe the general trends. The first notable pattern is the distinct seasonal rise and fall in beach width, corresponding very clearly to the winter and summer patterns of the year. Beach erosion in California is very correlated with this seasonal change, as the increases in storm and wave events in the winter months erode beaches, and the relatively calm conditions of the summer allow for sand to accumulate on the beach. But what are the localized dynamics of this system? And how do the width dynamics of the beach change throughout the site?



Beach width time series, average between all transects for a given date

Pulling and cleaning weather data

My hypothesis is that the seasonal pattern of the beach width timeseries is actually representing an increase in the frequency and intensity of storms during this season, and therefore I wanted to derive a timeseries of storm events for my date range, to study the influence of storm driven events on the beach morphodynamics.

Project purpose

When I initially searched for a comprehensive list of storm events for the Samoa region, I was not able to find a clean dataset, and many of the sources I did find focused on storms that would not have a significant impact on beach width (intense precipitation and flooding on land). Therefore for this project I set out to develop a KMeans clustering algorithm that would be able to parse through raw weather data from stations around Samoa beach, and create a timeseries of significant storm events.

KMeans clustering analysis of storm days

In order to find the most relevant data that would contribute to a holistic indicator of a storm event, I considered a number of weather variables commonly used to measure storm activity, and focused on only the variables I believed would influence beach width the most. I pulled wind, temperature, and mean sea level pressure data from the local airport, which conveniently lies just inland of Samoa beach. I also gathered precipitation data from the Global Historical Climatology Network, and wave height data from a buoy just offshore of Samoa beach. The wind and wave data is especially important when considering impacts of storm driven erosion of the coastline, temperature data is a great indicator of seasonality, and mean sea level pressure is an essential factor when considering the intensity of storms and has long been used as a weather indicator (barometer). I have compiled my data sources into the following table, along with their units (and the conversions I made) as well as their original organization which distributes the data.

Variable	Description	Units	Temporal Resolution	Source
WVHT_mean	Significant wave height (daily mean)	Meters (m)	Hourly converted to daily averages	NDBC Buoy Data (Humboldt bay entrance)
Daily_max_sust_knots	Maximum sustained wind speed	Knots (kt)	Hourly converted to daily maximum	NOAA ASOS (Arcata Airport -ACV station)
daily_mean_sust_knots	Mean sustained wind speed	Knots (kts)	Hourly converted to daily averages	NOAA ASOS
daily_min_press_hpa	Minimum sea-level pressure	Millibars	Hourly converted to daily minimum	NOAA ASOS
precip_mm	Total daily precipitation	Millimeters (mm/day)	Daily	NOAA GHCN-Daily (USW00024213)
daily_mean_tmpf_F	Mean daily air temperature	Fahrenheit (F)	Hourly converted to daily averages	NOAA ASOS

Data cleaning and preprocessing

Most of these datasets were on different timescales, and had different conventions when considering NaN data, as well as different datatype formats, so a lot of data cleaning was required in order to merge this weather data into a usable format. First I downloaded the precipitation data from the GHCN website. Ultimately it was easiest to query the url within the script, as there were many downloadable links and I was able to find out how to do this on stack overflow with the station ID and start and end date variables. The precipitation data was originally in tenths of a millimeter, requiring me to multiply by 10 to get total rain per day in millimeters. Next the wind, pressure, and temperature data from the local Arcata airport weather station, run by NOAA. This data came very clean from NOAA, and all I had to do was filter for my columns of interest and my date range, as well as convert the data to numeric columns, as it was originally stored as strings. I then converted each hourly timestamp and grouped by date into daily timestamps. For the pressure I grouped by daily minimum, as low pressure proceeds and indicates a

storm. For the wind I grouped into two new columns: daily maximum wind speed, and the daily average. The daily max gives information about the severity of the storm, and the daily average tracks wind speed more accurately across time. Finally for the temperature data I simply took daily averages in degrees fahrenheit. The wave dataset from the NDBC buoy data was a bit more convoluted, as some data errors and naming conventions were at first hard to parse. The data could not be queried in long form raw data, and instead had to be downloaded in individual years as zipped text files. I gathered the zipped text files into a single folder and iterated over them to unzip and read with `pd.read_csv` (which handled the space delimited text files well). I also removed a `#` from the year column that I thought may have been a mistake in some of the files and built my datetime column using the individual time columns. 'MM' was used throughout the files in place of a NaN marker, but `pd.read_csv` had a `na_values` argument to handle these. I concatenated all the files into a single wave height dataframe and filtered by year again to be sure my data fit in my timeseries. Finally, I resampled the hourly wave data into average daily wave height values, to fit the timesteps of my other weather data, and resampled and renamed my raw 'time' column to 'date'. I merged all the dataframes into a single `weather_df` dataframe by date using the `how='outer'` argument to merge all the data, and not just the days that perfectly overlapped (I wanted to keep as much data as possible in this step).

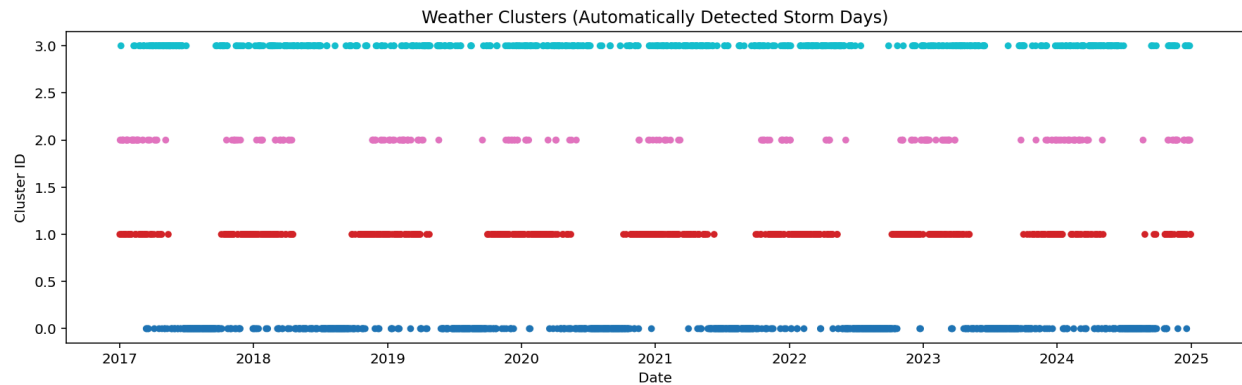
Clustering of transects analysis

Once the data was cleaned and merged into a single dataframe, I proceeded to build my KMeans clustering analysis of the dataframe to parse out dates of storms. I first made sure the 'date' column was a datetime object and sorted by date and dropped NaN values. I then created a features variable to store the column names of my features (redundant but I wanted to try to eliminate features or add others dynamically). I then created a feature array `X` and standardized it with the `sklearn.preprocessing.StandardScaler` to scale my data, avoiding over reliance on variables with large values, i.e pressure which is orders of magnitude greater than the other units. I looped through multiple `k` values for my model to see which silhouette score was highest. This resulted in an optimal `k` value of 2, but after playing around with the outputs and plots, I realized that this binary clustering was over emphasizing the precipitation values and that a `k` value of 4 identified the best storms. I confirmed this by printing the statistics in each cluster, and found that cluster 2 represented storm conditions better than all

	daily_max_sust_knots	daily_mean_sust_knots	daily_min_press_hpa
cluster			
0	10.42644	3.86023	1014.83319
1	8.84768	3.35211	1020.11960
2	18.42188	8.71667	1006.00156
3	16.10848	6.42174	1016.61474

	precip_mm	daily_mean_tmpf_F	WVHT_mean
cluster			
0	0.78266	56.00596	1.66604
1	0.86954	46.49797	2.37269
2	20.02070	50.78138	3.45106
3	1.80876	50.77160	2.75267

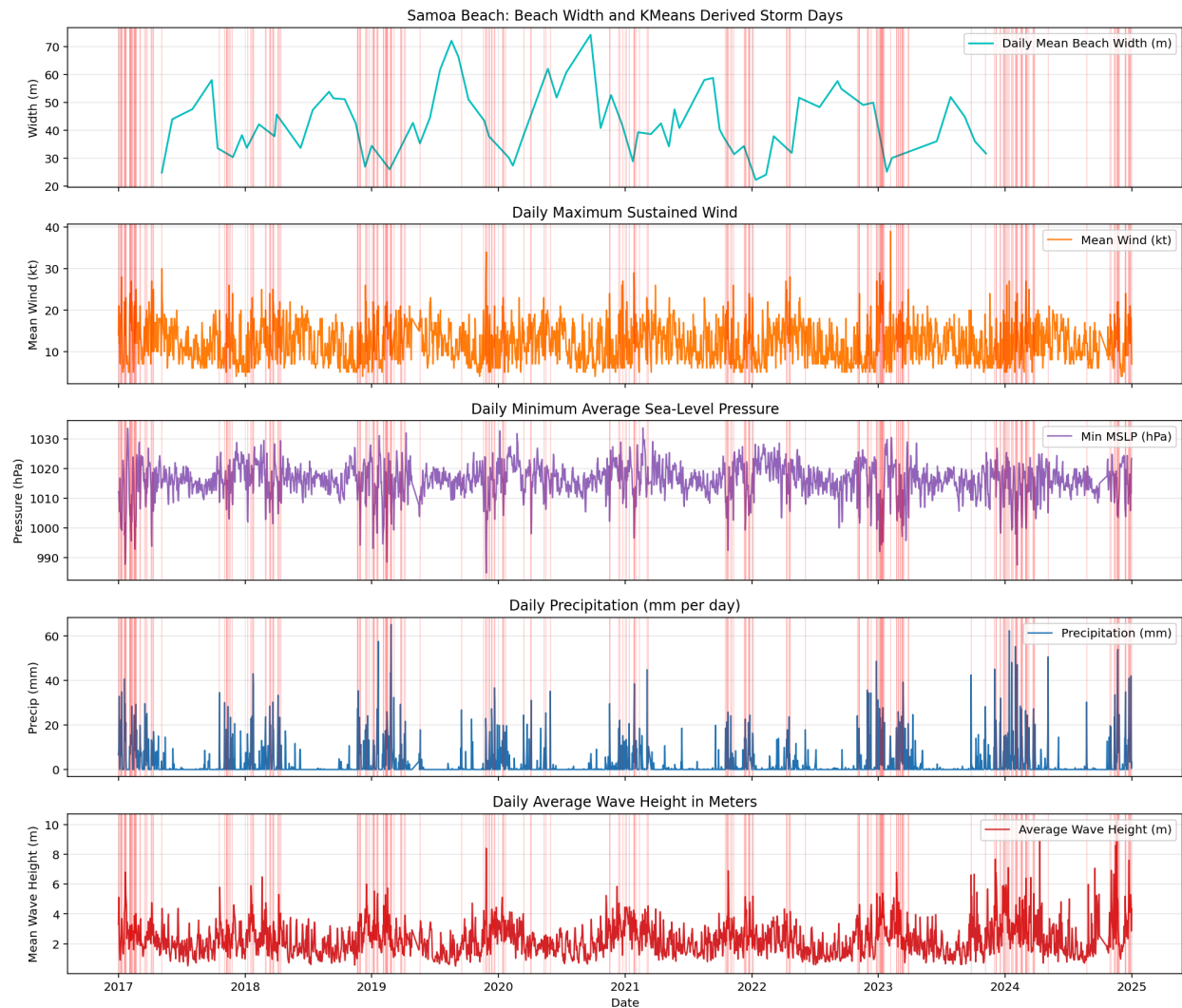
the others. In order to get this to print correctly I added a few arguments to control the output, since spyder (the ide I was using) did not seem to want to print all the columns. I then plotted the scatterplots of the cluster results per date to see, along with the summary statistics, which cluster was temporally representing the winter driven storm dynamics of the area. As can be seen, I chose cluster 2 as the representative storm classification, with its high max and mean wind speeds, low pressure, high precipitation, and relatively high wave heights. Cluster 2 also clearly showed the expected temporal



pattern of winter storms.

Visualizing storm dynamics with beach width

To plot my final KMeans identified storms throughout the time series and explore their interaction with the beach width timeseries, I created a five panel plot which showed the averaged beach width (over all transects) overlaid with the storm dates in red. I also included the individual weather data plots to see the interactions between the various elements and the beach width.



Analysis

From these plots there is a clear connection between storm activity and the erosion of Samoa beach, the dense clusters of storm activity in the winters drive this downturn in width and their subsidence allows the beach to swell as it gathers sand. The overlaid storm dates match well with large wave and wind events in the raw weather data, and correspond directly to spikes in the various weather timeseries. I tested the difference between setting a manually derived threshold for storm conditions, such as filtering for low pressure and high winds, high waves, and precipitation. But the thresholds I was able to find online were varied, and changed by region. This machine learning approach is a stable methodology which only took into account the local dynamics of the raw weather data I took directly around the site, and more aptly classified what was a ‘storm’ in Humboldt county. I will use this method to expand this analysis to all the

other regions where there are sandy beaches, and where manual thresholds would have to be tuned to specific regional weather standards.